

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (EC) Theory Examination

March-Apr 2018

EC202.01/EC242 Network Theory

Date: 30/03/2018, Friday

Time: 01:30 pm To 04:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

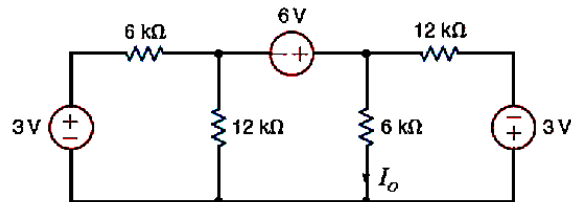
SECTION – I

Q:1 Answer the following [07]

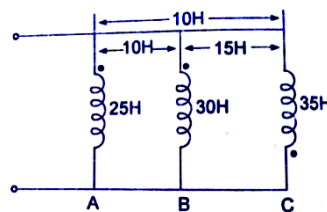
- a. State the difference between circuit and network. [02]
- b. Explain different types of network. [05]

Q:2 Attempt any three [18]

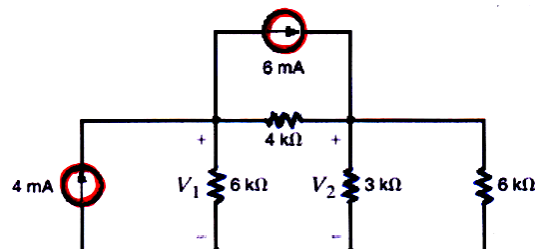
- a. Find I_o in the network using mesh analysis.



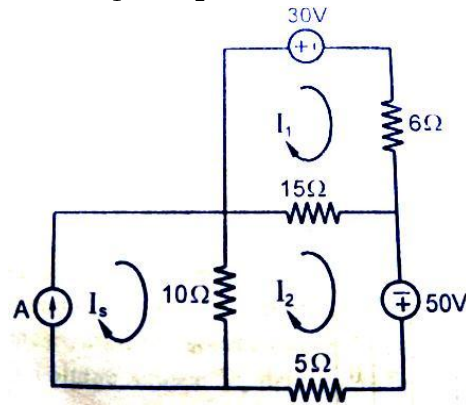
- b. Determine the inductance between the terminals for a three coil shown in following figure.



- c. Using node analysis find V_1 and V_2 in the circuit.



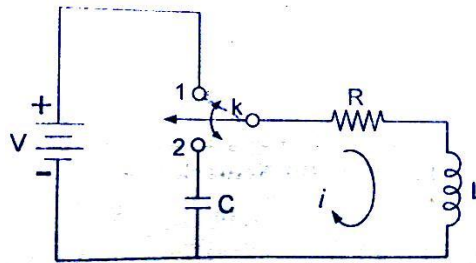
- d. Determine the mesh currents I_1 and I_2 in the network of following figure.



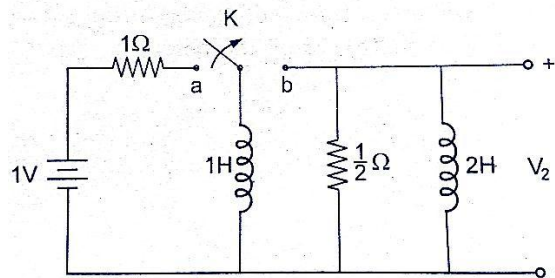
Q:3 Attempt any two

[10]

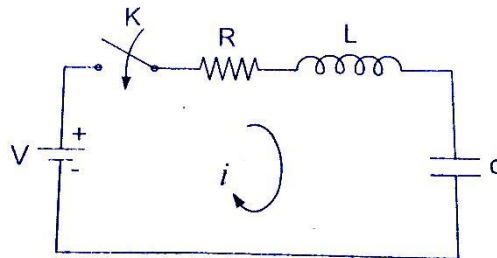
- a. In the network of following figure, switch K is changed from 1 to 2 at $t=0$. Find the values of i , di/dt and d^2i/dt^2 at $t=0^+$. Here, $R=1000\Omega$, $L=1H$, $C=0.1\mu F$ and $V=100V$.



- b. In the network of following circuit, the switch k is in position a for a long period of time. At $t=0$, the switch is moved from a to b. Find $V_2(t)$.



- c. In the network of following figure, the switch K is closed at $t=0$ with capacitor uncharged and with zero current in the inductor. Find the values of i , di/dt and d^2i/dt^2 at $t=0^+$. Here, $V=100V$, $R=10\Omega$, $L=1H$ and $C=10\mu F$.

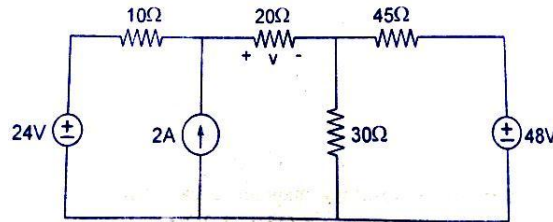


SECTION – II

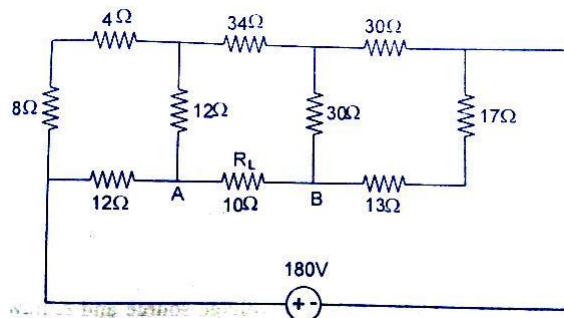
Q:4 Obtain the particular solution for R-C transient circuit. **[07]**

Q:5 **Attempt any three.** **[18]**

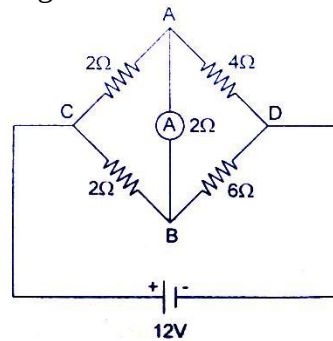
- a. Explain and analyze the response of series R-L-C circuit as related to the S plane location of roots.
- b. Use superposition theorem to find value of 'V' across 20Ω resistor in the following network.



- c. Find the current in 10Ω resistor in the network of following figure using thevenin's theorem.

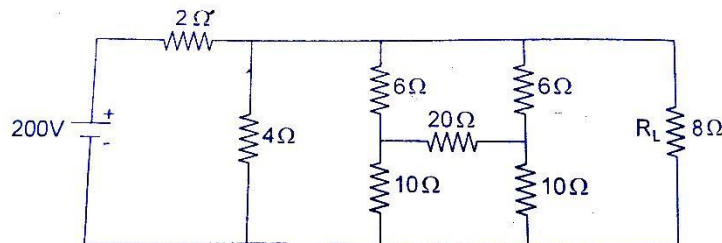


- d. Determine the current through ammeter of resistance 2Ω connected in the wheastone bridge of following circuit.

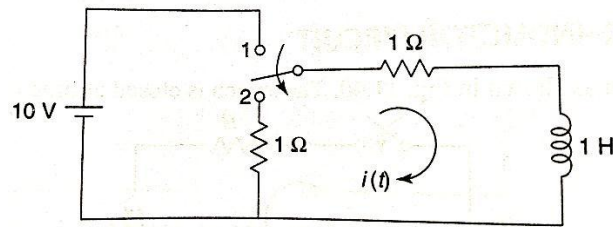


Q:6 **Attempt any two.** **[10]**

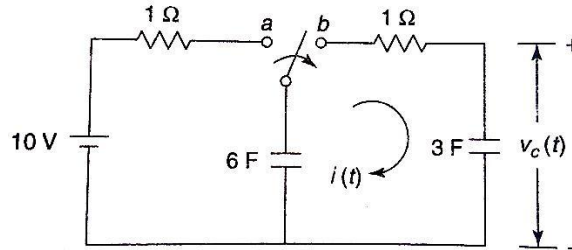
- a. Find the current through 8Ω resistor using Norton's theorem.



- b. In the network of following figure, the switch K is moved from position 1 to position 2 at $t=0$ (a steady state existing in position 1). Solve for the current $i(t)$ for $t>0$ using laplace transformation method



- c. In the following network, the switch is moved from a to b at $t=0$. Using laplace transform, determine $i(t)$ and $v_c(t)$



- d. For the network shown in following figure, write down the tie-set matrix and obtain the equilibrium equation in matrix form using KVL. Calculate the loop currents and branch currents.

